

Statistical Inference on Global Health Data: A comprehensive analysis of the WHO Life Expectancy Dataset

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June 24, 2026

Contents

1	Introduction	1
2	Dataset Description	1
2.1	Source and Scope	1
2.2	Variables and Definitions	1
2.3	Data Cleaning and Preprocessing	2
3	Methodology	2
3.1	Exploratory Data Analysis (EDA) Procedures	2
3.2	Parametric Statistical Methods	3
3.2.1	Welch’s Two-Sample T-Test	3
3.2.2	One-Way Analysis of Variance (ANOVA)	3
3.2.3	Multiple Linear Regression (MLR)	3
3.3	Diagnostic tests	4
3.4	Non-parametric Statistical Methods	4
3.4.1	Mann-Whitney U Test (Wilcoxon Rank-Sum Test)	4
3.4.2	Kruskal-Wallis Test	4
3.4.3	Spearman Rank Correlation	4
4	Results	5
4.1	Exploratory Data Analysis (EDA) procedures	5
4.2	Parametric Statistical Methods	8
4.2.1	Welch’s Two-Sample T-Test	

1 Introduction

Throughout history, man has desired to increase his lifespan. This is seen from the different advancements in technology, medicine as well as nutrition across the centuries. Man has tried to improve his survival as this deeply woven in his DNA. From the discovery of fire, to the invention of the wheel and to the discovery of penicillin, man has had an innate desire to improve his life expectancy. Looking at the modern era, this is till the case albeit in a more complex manner. Several factors today contribute to the life expectancy of humanity. These factors range from the environment whose conservancy is a very hot topic in the modern era to the socio-economic status of a given country. Not a single factor can single handedly determine the life expectancy of a given country. The World Health Organization (WHO) has been at the forefront in analyzing health data which very much coincides with life expectancy. These data are very important especially in this modern era as it aids in policy making for governments and other stakeholders. For instance, the Government of Kenya in 2010 through the Republic of Kenya (2010), enacted The Alcoholic Drinks Control Act, commonly referred to as the 'Mututho Law'. This was a with a view of reducing the consumption of alcohol which had plagued the country for a while.

The study selected the **WHO Life Expectancy Dataset** (Gochiashvili, 2023) for its comprehensive nature and relevance to global health. The dataset contains 2,864 observations and 21 variables. This includes both numerical and categorical data such as country, region, life expectancy, adult mortality, alcohol consumption, gross domestic product (GDP) on health and more. This diversity of the variables allowed for analysis of the factors influencing life expectancy across different countries and regions. The project aimed to model the predictors of global life expectancy and evaluate geographical and economic inequalities. The analysis applied various statistical inference methods. This included both parametric methods and their non-parametric counterparts when assumptions are violated. The statistical questions that were addressed were:

- Is there a significant difference in life expectancy between developed and developing countries?
- Are there variations in life expectancy across different regions of the world?
- What are the key predictors of life expectancy, and how do they interact with each other?
- How do the results from parametric methods compare to those from non-parametric methods when assumptions are not met?

Table 1: Description of key variables used in the analysis

Variable	Type	Description
Life_expectancy	Cont.	Life expectancy at birth (years). (Dependent Variable)
Economy_status	Cat.	Economic classification (1 = Developed, 0 = Developing).
Region	Cat.	Nine distinct global geographic zones.
Schooling	Cont.	Average years of educational attainment.
Alcohol	Cont.	Per capita consumption of pure alcohol (liters).
BMI	Cont.	Average Body Mass Index.
Polio	Cont.	Polio immunization coverage (%).
Incidents_HIV	Cont.	HIV incidents per 1,000 population.
Thinness	Cont.	Prevalence of thinness (10–19 years) (%).
GDP_per_capita	Cont.	GDP per capita in USD.
Population_mln	Cont.	Total national population (millions).

2.3 Data Cleaning and Preprocessing

Continuous variables were inspected by the use of histograms, scatter-plots and heatmaps. This aided in identifying the distribution of the data and the relationships between the variables. Variables such as GDP and Population in millions were log transformed due to their skewness. High linear associations were indeed flagged and omitted as this would hurt the validity of the parametric tests that followed.

3.2 Parametric Statistical Methods

3.2.1 Welch's Two-Sample T-Test

The Welch Test kicked off the statistical inference of the data. The study's aim here was to investigate if there was a difference in life expectancy between Developed and Developing countries. The Welch Two sample T-test was selected in order to account for the potential unequal variance of the two groups. The test statistic is calculated as:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

where $\bar{X}_i$, s_i^2 , and n_i represent the sample mean, sample variance, and sample size of group i

3.3 Diagnostic tests

Diagnostic tests proceeded in order to validate the outcome of the parametric tests. This comprised a series of visualizations which elaborated if the assumptions were held or indeed violated. The assumptions tested were as follows:

- **Linearity:** Done via *Residuals vs. Fitted* plots to check for random scattering. Structural patterns indicate non-linear relationships.
- **Homoskedasticity:** Evaluated using *Residuals vs. Fitted* and *Scale-Location* plots. Constant vertical spread across all fitted values confirms stable variance. Unequal indicates heteroskedasticity.
- **Normality of Errors:** Inspected via *Normal Q-Q* plots. Adherence of standardized residuals to the 45° reference line confirms normality as systematic deviations signal heavy tails or severe skewness.
- **Absence of Multicollinearity:** Quantified using the Variance Inflation Factor (VIF) for each predictor:

$$\text{VIF}_k = \frac{1}{1 - R_k^2}$$

A conservative threshold of $\text{VIF} > 5$ was established to flag and isolate problematic correlated variables.

4 Results

4.1 Exploratory Data Analysis (EDA) procedures

Initial EDA's evaluated the distributions as well as the descriptive statistics of the $N = 2,864$ observations. The study obtained the summary statistics of the numeric variables. Table 2 exhibits the various measures of central tendency and dispersion.

Table 2: Descriptive Statistics of the Global Health Dataset

Variable	Mean	Median	Variance	SD	Range	IQR
Infant Deaths	30.36	19.60	758.35	27.54	136.30	39.25
Under-Five Deaths	42.94	23.10	1,986.48	44.57	222.60	56.33
Adult Mortality	192.25	163.84	13,204.37	114.91	669.98	139.88

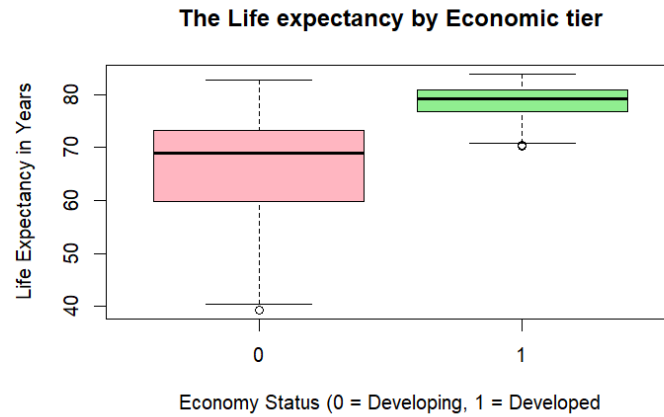
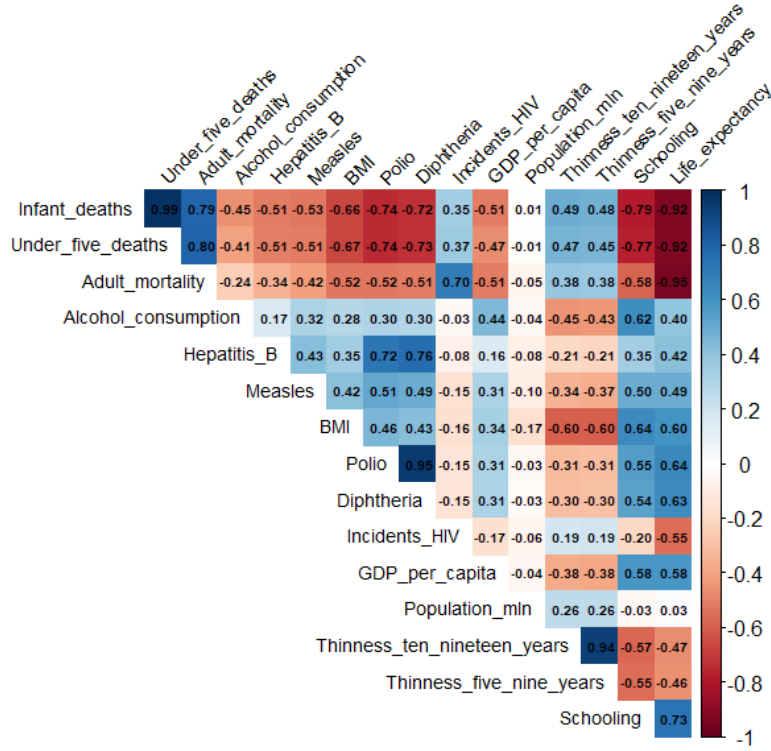
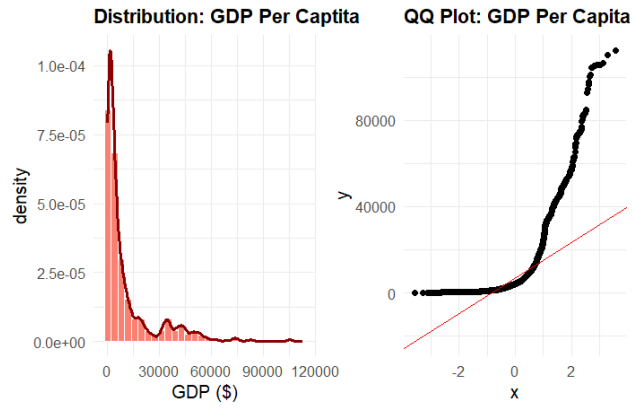


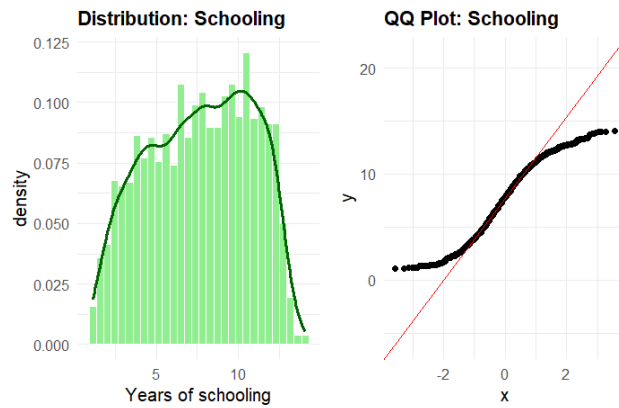
Figure 2: Comparison of life expectancy distributions across distinct economic tiers, where 0 represents developing nations and 1 denotes developed nations. Developed countries display a significantly higher median life expectancy and a much narrower interquartile range, whereas developing nations exhibit far greater volatility and lower overall health outcomes.

The study investigated the relationship between life expectancy and schooling through a scatter-plot. Figure 3 shows the positive relationship between them whereby as years of schooling increases,





(b) Distribution and Q-Q plot for GDP per Capita



(c) Distribution and Q-Q plot for Schooling

Figure 5: Parametric diagnostics (Part II). The left panels display empirical density curves against standard histograms, while the right panels represent normal Q-Q plots.

4.2 Parametric Statistical Methods

4.2.1 Welch's Two-Sample T

4.2.2 One-Way Analysis of Variance (ANOVA) and Tukey post-hoc test

A One-Way ANOVA evaluated the variation in life expectancy across the nine global geographic regions. The analysis produced a p-value of $p = [2.2\text{e-}16]$. The Tukey post-hoc test compares multiple regions together. **North America - Africa** for instance had a difference of 19.99 years with a p-value < 0.0001 . Table 3 shows the Tukey post-hoc results.

Table 3: Tukey HSD Post Hoc Pairwise Comparisons for Life Expectancy by Geographic Region

Comparison (Region Pair)	Difference	Lower CI (95%)	Upper CI (95%)	p-adj
Asia - Africa	11.608	10.537	12.678	< 0.0001
Central America and Caribbean - Africa	14.590	13.381	15.799	< 0.0001
European Union - Africa	19.8			

4.2.3 Multiple Linear Regression (MLR)

This was the last parametric test. It evaluated the extent by which the economic indicators, healthcare of a country, physical metrics and demographic scales such as region predict the life expectancy of a country. Multi-collinearity was checked via the VIF scores as a threshold of > 5 was set. **Region** and **Schooling** had a VIF of 18.623 and 5.075 respectively but were both retained. Table 4 shows the multiple regression model denoted as **healthy model**. A bi-directional stepwise optimization was done to obtain the optimum model and table 4 exhibits it. The adjusted R squared was 0.891. A comparison of both the multiple regression and the optimum model is shown below:

Table 4: OLS Regression Comparison: Baseline MLR vs. Optimum Stepwise Model

	<i>Dependent variable: Life Expectancy (Years)</i>	
	Healthy Baseline Model (1)	Optimum Stepwise Model (2)
<i>Geographic Region (Baseline: Africa)</i>		

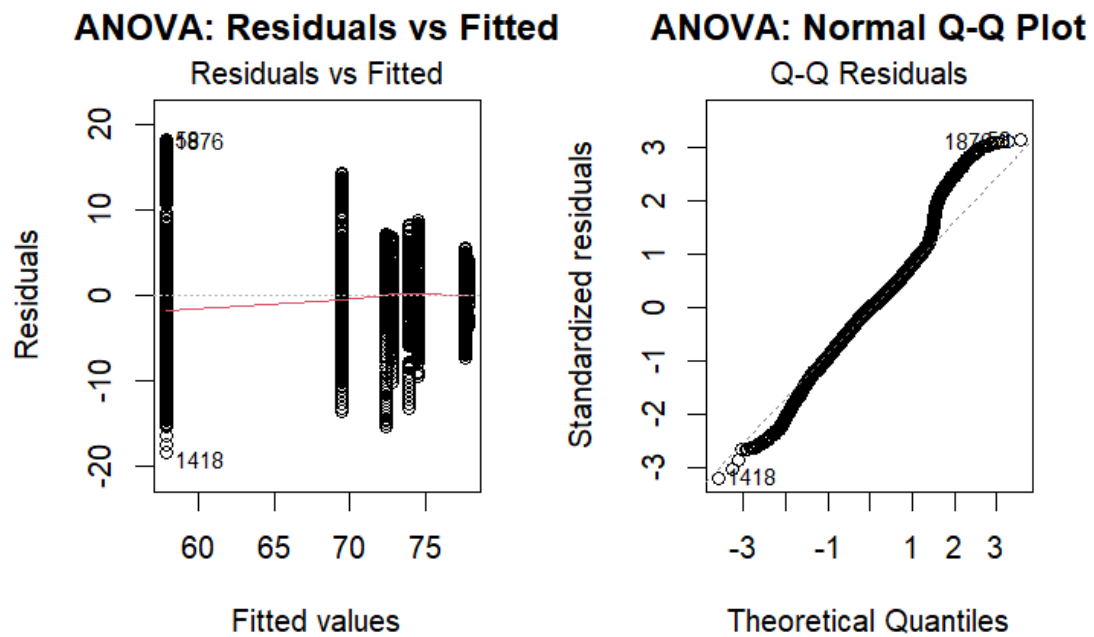
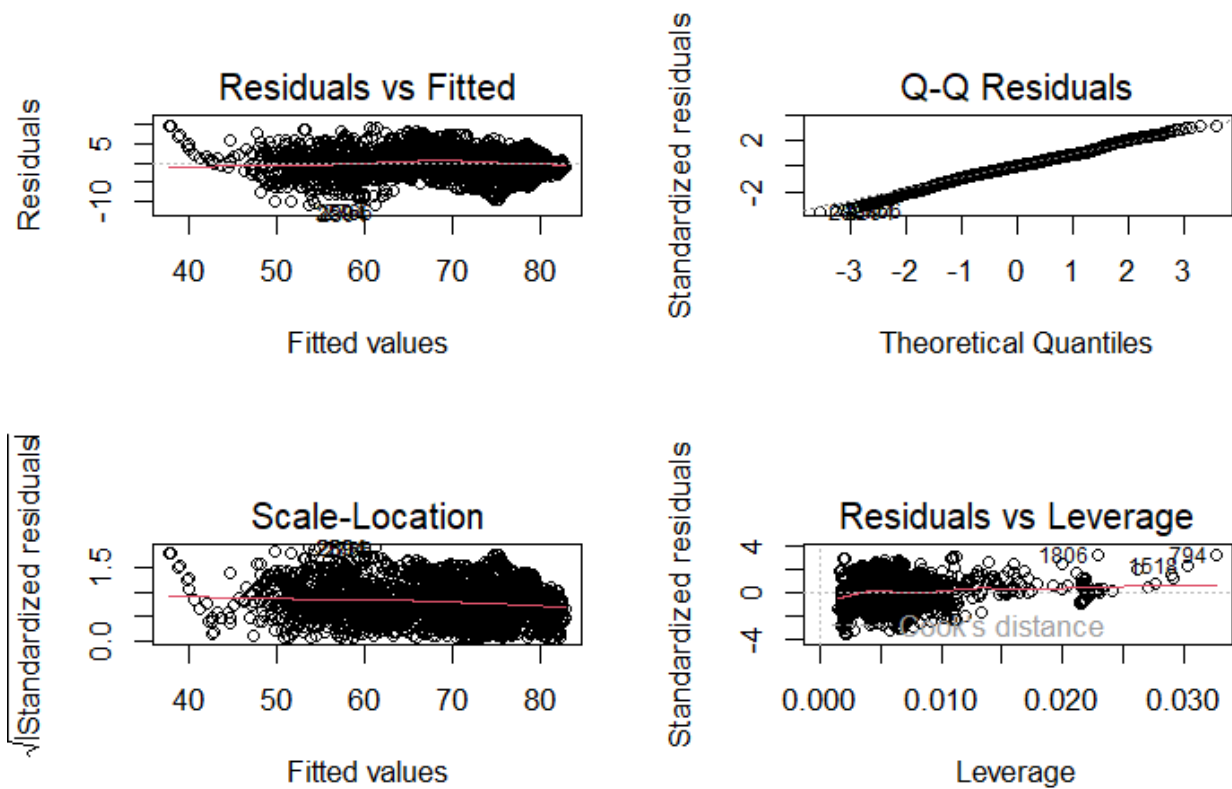


Figure 7: ANOVA Diagnostic Plots: Residuals vs Fitted and Normal Q-Q



4.4 Non-parametric Statistical Methods

4.4.1 Mann-Whitney U Test (Wilcoxon Rank-Sum Test)

The Wilcoxon Rank-Sum test investigated whether there was a significant difference in the distribution of life expectancy between developed and developing nations. It yielded a test statistic of $W = [85918]$, with a resulting p-value of $p = [2.2\text{e-}16]$.

4.4.2 Kruskal-Wallis Test

Kruskal-Wallis test evaluated if there was a significant difference in distribution of life expectancy across different geographical regions in the world. The test returned a chi-squared statistic of $\chi^2 = [1678.9]$, with degrees of freedom $df = [8]$ and a p-value of $p = [2.2\text{e-}16]$.

4.4.3 Spearman Rank Correlation

Spearman Rank correlation checked if there was a monotonic relationship between years of schooling and life expectancy. The calculation resulted in a correlation coefficient of $\rho_s = [0.74]$, accompanied by a p-value of p

GDP_per_capita. A strong negative relationship was exhibited between **Life_expectancy** and **Adult_mortality**. This inverse relationship can be explained as the life expectancy increases, it results in a decrease in the rates of adult mortality. Several variables had a close to perfect relationship. These were **Polio** and **Diphtheria**, **Infant_deaths** and **Under_five_deaths**, and **Thinness_ten_nineteen_years** and **Thinness_five_nine_years**. They were omitted due to them being smoking guns for multi-collinearity, in order to maintain statistical accuracy in the subsequent parametric tests. To crown the EDA section, assessment of the distributions was done. This was done onto our dependent variable **Life_expectancy**, **GDP_per_capita** and **Schooling**. GDP was selected due to its already established prevalent skewness as **Schooling** was selected due to its median and mean values being close together. Density plots and Q-Q plots were constructed for each. For all three variables, their Q-Q plots had heavy tails which meant that they were skewed. This was seconded by their Shapiro-Wilk test whose p-values were well below 0.05 indicating their significance and violation of normality.

5.2 Statistical Analysis

5.2.1 Economic and Geographic differences

With guidance

dropped.

- **Log_GDP** and **Economy_status_Developed** had VIF scores greater than 5. Since they both explained the same thing in that a country with a high GDP is in a higher economic tier, we dropped the categorical variable **Economic_status_Developed**.
- **Region** had a VIF score of 18.6. It was not omitted due its properties. The dataset had 9 different regions making the Degrees of Freedom to be 8. A series of dummy variables were created which in turned inflated the VIF. The GVIF penalized it due to the many dummy variables so that it could be observed on an equal mathematically playing field. Squaring it (1.2) leads to ≈ 1.4 , which was much below the 5 threshold.
- **Schooling** had a VIF score of 5.07. As explained by Grossman (1972) and Lleras-Muney (2005), dropping **Schooling** would introduce severe omitted variable bias. This is because it is a fundamental structural determinant of health. Additionally, Balaj et al. (2024) argues that every single additional year of schooling reduces the risk of adult mortality by 1.9%. Finally, a VIF of 5 falls well within acceptable limits according to standard econometric practices as explained by O'Brien (2007). Its exclusion would misrepresent the independent causal effect that educational attainment exerts on life expectancy separate from income. We decided not to drop **Schooling** as it was essential for this project.

following were the findings:

- **Homoskedasticity** - Both the Residual vs Fitted plot and the Scale-Location plot showed uneven spread of residuals across the fitted values. This indicated heteroskedasticity.
- **Normality** - Normality was violated as exhibited by the QQ Plots. There were heavy tails thus visually presenting this violation.
- **Linearity** - From the Residuals vs Fitted plot, the red line showed a curve rather than a red straight line. This proved that linearity was violated.
- **Influential Outliers** - The Residuals vs Leverage plot showed that there were no points outside the cook's distance. This informed us that there were no influential outliers in the dataset.

From the diagnostics, it was seen that most of the assumptions were violated thus a relaxation of the assumptions was needed through the employment of non-parametric ones. Due to non-parametric tests reliance on median rather than mean, they effortlessly bypassed the observed skewness and unequal variances. The fact that the non-parametric p-values ($p < 0.05$) perfectly aligned with the significance levels of the parametric tests solidifies the conclusions of this study. The identified economic and geographic drivers of life expectancy are highly robust and exist independently of the specific statistical distribution of the dataset.

5.3 Practical Implications

6.2 Statistical Conclusions

Our framework successfully validated the core objectives of the study. The parametric tests being our baseline models laid the foundations of the project although diagnostic visualizations revealed the violations of normality and homoskedasticity. Our transition to non-parametric tests confirmed the earlier observed economic and geographic differences in life expectancy. They proved that these disparities were indeed genuine and not plagued by influential outliers.

6.3 Limitations of the Study

We experienced one major limitation during the analysis of this dataset. Due to the various correlation metrics that were used in the study, from heatmaps to VIF scores, several variables were omitted. This was to prevent our models from crashing due to high multicollinearity. This led to some dynamics and nuances being lost. For instance, `Under_five_deaths` and `Adult_mortality` were omitted for having high VIF scores as they were directly used in the calculation of life expectancy. Some variation was possibly lost due to this. This showed that purely linear models may oversimplify the complex nature of global health data.

6.4 Recommendations for Future Research

Future studies on global health data especially life expectancy should transition to time series analysis. This is due to the yearly changes of GDP in several countries which would in turn (if positive) improve other sectors such as vaccination coverage or schooling. Future studies should also focus on employing machine learning models such as Random